Digital Image Processing Lab (NCSH 302) **Session: Monsoon 2026-27**
B. Tech. 5th Sem (CSE-Hons) **Date: 18.08.2026**

Assignment-2 (Pixel Geometry-II & Image Enhancement-I)

1. Let $Im_{binary}(x, y)$ be a binary image and let *S* denote the set of foreground (1-valued) pixels in it. Suppose *S* consists of *K* disjoint connected components (regions) $R_1, R_2, \dots, R_K$, obtained using a specified connectivity $con(4 - or\ 8 - connectivity)$. Two regions $R_i\ and\ R_j$ are said to be adjacent if their union $R_i \cup R_j$ forms a connected set; otherwise, they are disjoint. The boundary of a region *R* is the set of pixels in *R* that have one or more neighbors not in *R*. Given a binary image $Im_{binary}(x, y)$ containing its *K* regions, write the following functions:
   i. adj_{flag} = Region_adjacency $(Im_{binary},\ R_i,\ R_j)$ – returns 1 if regions R_i and R_j are adjacent, and 0 if they are disjoint.
   ii. $Im_{boundary}$ = Boundary_extract (Im_{binary}, con) – extracts the boundary pixels of every region present in Im_{binary} using the connectivity *con* and reports the boundary pixel count separately for each region.
   iii. Im_{fg}, Im_{bg} = Foreground_background (Im_{binary}) – separates and returns the foreground image (union of all regions) and the background image (its complement) and reports the total pixel count of each.

Sample Input:

Binary Image (8 × 8):

0	0	0	0	0	0	0	0
0	1	1	0	0	0	0	0
0	1	1	0	0	1	1	0
0	0	0	0	1	1	0	0
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

Sample Input:

Number of connected regions = 3
Region 1 = 4 pixels
Region 2 = 4 pixels
Region 3 = 4 pixels

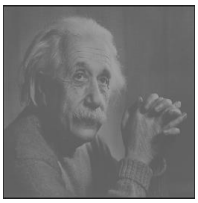
con = 4-connectivity
$Region_adjacency\ (R_1, R_2)$ = 0

Boundary pixel count:
Region 1 = 4
Region 2 = 4
Region 3 = 4

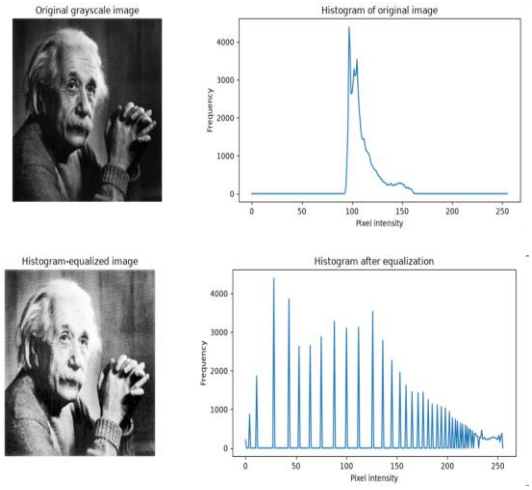
Foreground pixel count = 12
Background pixel count = 52

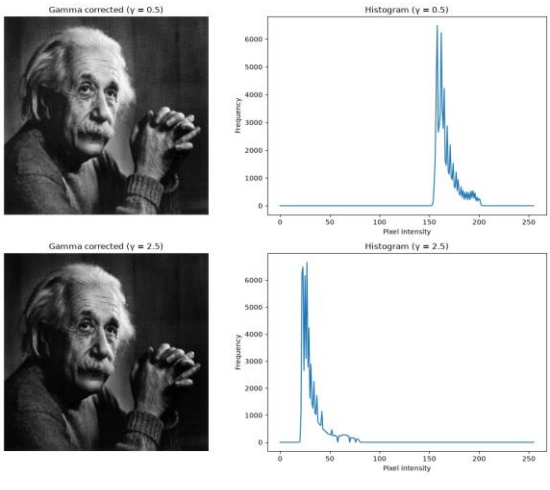
2. A 256×256 grayscale image with $8 - bit$ pixel depth is given, in which the image has low contrast and some details are difficult to distinguish. To improve the contrast, two enhancement methods are considered: histogram equalization and gamma correction. Read the given image "input.png", plot its histogram, and write a function $hist_eq\ (Im)$ to perform histogram equalization and display the enhanced image. Also, write a function $gamma_correction\ (Im, gamma)$ and apply gamma correction using $\gamma = 0.5\ and\ \gamma = 2.5$. Display the original image and histogram, the histogram-equalized image and histogram, and the two gamma-corrected images with their corresponding histograms.
[Hint: Plotting histogram in OpenCV-
import cv2
import numpy as np
from matplotlib import pyplot as plt
img = cv2.imread('pout.tif',0)
cv2.imshow("image", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
hist = cv2.calcHist([img],[0], None, [256],[0,255]) plt.plot(hist)]

Sample Input:



Sample Output:





3. A $256 \times 256, 8-bit$ grayscale image "lake.png" with uneven illumination is given. Perform global thresholding on the entire image Im using $T_{global}(Im) = \frac{1}{3} \times [max(Im) - min(Im)]$, and then divide the image into 64 non-overlapping 32×32 regions R_i, calculate the local threshold for each region using $T_{local}(R_i) = \frac{1}{3} \times [max(R_i) - min(R_i)]$, and perform local thresholding. Print the global and all 64 local threshold values, display the globally and locally thresholded binary images, and display the difference image obtained by subtracting the global thresholded image from the local thresholded image.

Sample Input:



Sample Output:

Global threshold = 74.66
Local Thresholds:
$Block_1$ = 47.33
$Block_2$ = 60.33
$Block_3$ = 61.67 … similarly for all the blocks

